Effects of Feedback on Students' Motor Skill Learning in Physical Education

Does feedback help students learn movement skills—
and which kinds help most?

Yulan Zhou · Wei De Shao · Lijuan Wang

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Smart Chen & Xinyue Xia

Why motor skill learning matters in school PE

Participation in physical activity

Children with better motor skills are more likely to participate in physical activity.

Links to physical fitness

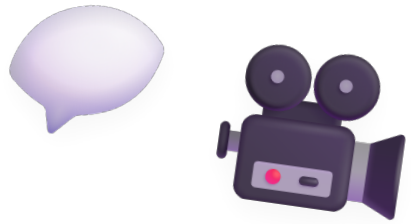
Motor skills are associated with cardiorespiratory fitness, muscular endurance, and strength.

School PE as a learning setting

School PE offers opportunities to develop motor skills, a central goal of the curriculum.

Which teaching strategies help students learn them? **This review focuses on feedback.**

Does feedback help?



Mixed findings

Some studies found improved motor skills.
Others found no significant difference.

Different feedback designs

Spoken guidance, video replay,
praise, and corrective comments offer different kinds of information.

The review brings these studies together to assess the strength of the evidence.

Research questions

1 Does feedback improve motor skill learning in school PE compared with no feedback?

2 How do feedback format and content affect learning?

What is feedback?

Information about a learner's performance that may help them **identify errors, adjust movements, and keep practising.**

Provider · Who gives it?

Teacher, peer, or video system

A single message can come from one provider or a combination.

Format · How is it delivered?

Spoken, written, gestural, or visual

Video replay is one example of visual feedback.

Content · What does it say?

Praise

"Good effort."

Movement information

"Your elbow was below shoulder height."

Correction

"Raise your elbow on the next attempt."

One message can combine **praise, movement information, and correction.**

Paper §§2.3, 4.2–4.3 · Messages are illustrative examples

Why might feedback needs differ?

Task demands

Movement

Discrete
A throw

Continuous
Swimming

Serial
A gymnastics routine



Perception
Open: changing conditions
Closed: stable conditions

Learner experience

Fitts & Posner's stages of skill learning

1 **Cognitive**
Understanding the movement

2 **Associative**
Refining technique and consistency

3 **Autonomous**
Performing with little conscious attention

Possible feedback needs
Beginners: instructions on how to move.
Experienced learners: specific cues about errors.

What this review adds

1	**PE-specific synthesis**	23 intervention studies of feedback in school PE classes across 11 countries.
2	**Element-level evidence**	Separates format and content; grades the evidence for each comparison.
3	**Guidance for teachers**	Give feedback regularly; combine praise with specific information rather than praise alone.
4	**A research agenda**	High-quality RCTs, feedback frequency, and technology-delivered video feedback.

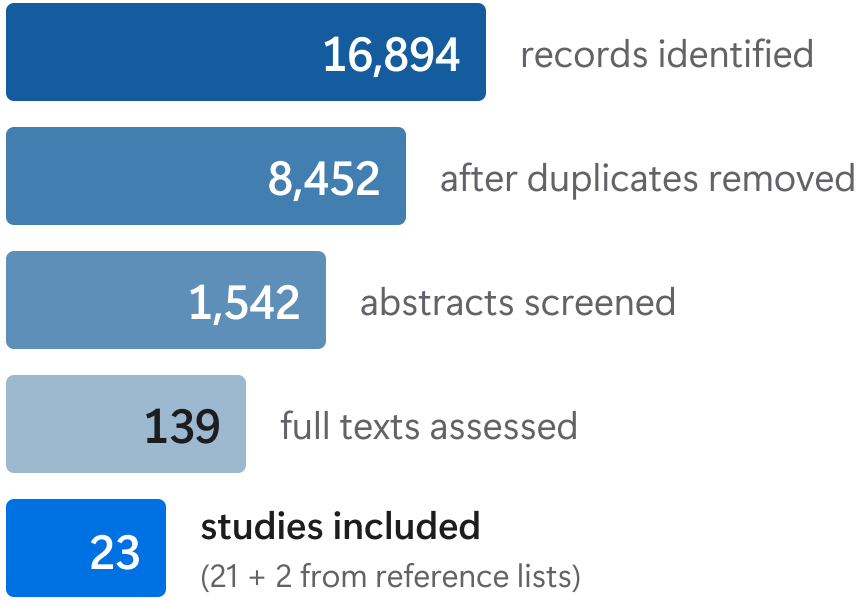
A PRISMA-guided search narrowed 16,894 records to 23

7 databases searched to April 2021: Academic Search Premier, ERIC, MEDLINE, Psych & Behavioral Sciences, Scopus, SPORTDiscus, Web of Science

Keywords: feedback × PE × motor skill

Included: feedback interventions in school PE, general K-12 students, peer-reviewed English full texts

Two independent reviewers, a third to resolve disagreements



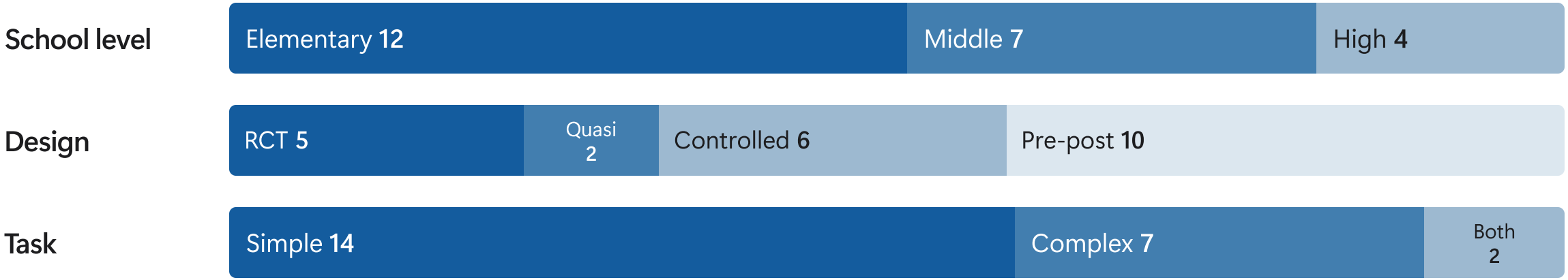
Bar lengths not to scale · Zhou, Shao & Wang (2021), Figure 1

Profile of the 23 included studies

23 studies, 1987–2019;
13 since 2010

11 countries;
10 studies from the USA

3–950
participants;
16 studies had fewer than 100

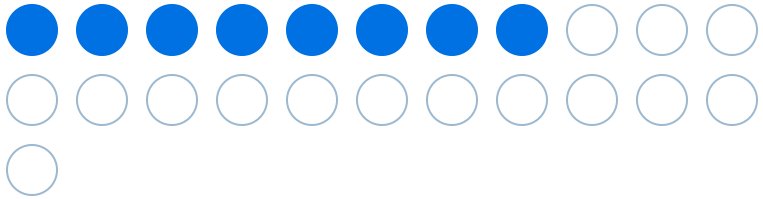


Zhou, Shao & Wang (2021), Section 3.2 and Table 1

Quality appraisal and evidence grading

PEDro scale

0–10; ≥ 5 = high quality



8 high quality (filled)
15 low quality (outline)
scores ranged 1–7

Weakest item: blinding.
18 of 23 studies scored 0.

Best-evidence synthesis (Slavin, 1995)

Level	Criterion
Strong	Consistent in ≥ 3 high-quality studies
Moderate	Consistent in 2 high-quality studies
Limited	Consistent in 1 high-quality study
Conflicting	Findings contradict each other
Insufficient	Consistent in low-quality studies only

Consistent = ≥ 2/3 of studies significant in the same direction.
No meta-analysis: feedback elements were too heterogeneous.

Zhou, Shao & Wang (2021), Sections 2.4–2.5 and Table 2

Evidence strength differs greatly by comparison

Comparison	Results	Evidence	XR design
Feedback	10 (76.9%) for 13 studies	**Strong**	Individual feedback
Visual vs verbal	3 (100%) for 3 studies	Limited	3D replay
Expert model vs self model	2 (66.7%) for 3 studies	Limited	Set experts
Praise + information vs praise	2 (66.7%) for 3 studies	**Moderate**	Give rewards
Information vs praise or corrective	4 (57.1%) for 7 studies	Conflicting	Adaptive feedback

Strengths

- Two independent reviewers for screening and rating
- Best-evidence synthesis fits heterogeneous designs
- Full study table makes each judgment traceable

Limitations

- 65.2% of studies low quality (PEDro < 5)
- 69.6% had fewer than 100 participants
- 78.3% lacked randomisation

Where could feedback support practice?

1 **Technical sports**

Learning golf, tennis, and other ball games



2 **Manufacturing**

Operating a machine with motion tracking



Questions?

Proposed applications · Effectiveness still needs testing